VITyarthi | Open Source Software

The Open Source Audit

A Capstone Project for OSS NGMC Course

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1. Introduction

Git is a version control system that's been around since 2005. It was created by Linus Torvalds, the same guy who made Linux. Git is used by developers to keep track of changes made to their code. It's basically a way to manage different versions of your code, so you can experiment with new features without messing up the original. Git is really popular, and most developers use it. It's free and open source, which means anyone can use it and contribute to it. Git has become the standard for version control, and it's used by big companies like Google and Microsoft. It's also used by individual developers who just want to keep track of their own projects. Git is pretty flexible, and it can be used for all sorts of projects, from small scripts to huge operating systems. One of the cool things about Git is that it's distributed, which means that every developer has a full copy of the entire project history. This makes it really easy to collaborate with other developers, because you can all work on the same project without having to be in the same place. Git also has a lot of tools and features that make it easy to use, like branching and merging. These features allow you to experiment with new ideas without affecting the main project. Git is also really good at tracking changes, so you can see exactly what's been changed and who changed it. This makes it really easy to collaborate with other developers, because you can see exactly what they've done. Git is widely used, and it's become the standard for version control. It's free, open source, and really flexible, which makes it a great tool for developers. Git has a lot of features that make it easy to use, and it's really good at tracking changes. It's also distributed, which makes it easy to collaborate with other developers. Overall, Git is a really useful tool that's widely used by developers. It's free, open source, and really flexible, which makes it a great choice for all sorts of projects.

Part A — Origin and Philosophy

A1. Problem this software solves

So what problem does Git solve? Well, before Git, developers used to have to keep track of their code manually. They'd make a copy of their code, rename it, and then make changes to the new copy. This was really time consuming, and it was easy to lose track of what changes you'd made. Git solves this problem by giving you a way to track changes automatically. You can make a branch, which is like a copy of your code, and then make changes to the branch. When you're happy with the changes, you can merge the branch back into the main project. This makes it really easy to experiment with new ideas without affecting the main project. Git also makes it easy to collaborate with other developers. You can all work on the same project, and Git will track all the changes you make. This makes it really easy to see what's been changed and who changed it. Git also has a lot of tools and features that make it easy to use, like pull requests and code reviews. These features allow you to review code before it's merged into the main project, which helps ensure that the code is good quality. Git is also really good at tracking changes, so you can see exactly what's been changed and who changed it. This makes it really easy to collaborate with other developers, because you can see exactly what they've done. Git is widely used, and it's become the standard for version control. It's free, open source, and really flexible, which makes it a great tool for developers. Git has a lot of features that make it easy to use, and it's really good at tracking changes. It's also distributed, which makes it easy to collaborate with other developers. Git is a really useful tool that's widely used by developers. It's free, open source, and really flexible, which makes it a great choice for all sorts of projects. Git has become the standard for version control, and it's used by big companies and individual developers. It's a great tool for collaborating with other developers, and it's really good at tracking changes.

A2. License Analysis

Git is open source, which means that anyone can use it and contribute to it. The open source license for Git is the GNU General Public License, which is a pretty standard license for open source software. This license allows anyone to use Git for free, and it also allows them to modify it and distribute their own versions. The license requires that any modifications to Git must also be distributed under the same license, which ensures that Git remains free and open source. This license is really important, because it ensures that Git remains a community-driven project. Anyone can contribute to Git, and anyone can use it for free. The open source license for Git is one of the reasons why it's become so popular. It's free, so anyone can use it, and it's open source, so anyone can contribute to it. The license is also really flexible, which makes it easy for developers to use Git in their own projects. Git is widely used, and it's become the standard for version control. It's free, open source, and really flexible, which makes it a great tool for developers. The open source license for Git is an important part of its success, and it's one of the reasons why Git has become so widely used. Git is a really useful tool that's widely used by developers, and its open source license is a big part of its appeal. The license allows anyone to use Git for free, and it also allows them to modify it and distribute their own versions. This makes Git a really flexible tool that can be used in all sorts of projects. Git is a great choice for developers who want a free and open source version control system. It's widely used, and it's become the standard for version control. The open source license for Git is an important part of its success, and it's one of the reasons why Git has become so widely used.

A3. Ethics of Open Source

The ethics and community values behind Git are really important. Git is a community-driven project, which means that it's maintained and developed by a community of volunteers. This community is really active, and it's made up of developers from all over the world. The community values behind Git are all about collaboration and

openness. The community believes in making Git free and open source, so that anyone can use it and contribute to it. The community also values transparency and accountability, which means that all changes to Git are tracked and visible to everyone. This makes it really easy to see what's been changed and who changed it. The community values behind Git are also about meritocracy, which means that the best ideas and contributions are recognized and rewarded. This creates a really collaborative and inclusive community, where anyone can contribute and make a difference. The ethics behind Git are also really important. The community believes in making Git accessible to everyone, regardless of their background or experience. This means that Git is designed to be really easy to use, even for developers who are new to version control. The community also believes in being respectful and inclusive, which means that everyone is treated with respect and dignity. This creates a really positive and supportive community, where anyone can feel welcome and included. Git is a really useful tool that's widely used by developers, and its community values and ethics are a big part of its appeal. The community values behind Git are all about collaboration and openness, and the ethics behind Git are all about making it accessible and inclusive to everyone.

Part B — Linux Footprint

Git has a really big footprint in the Linux world. It's used by most Linux developers to manage their code, and it's become the standard for version control in the Linux community. Git is also used by many other open source projects, including the Linux kernel itself. This means that Git is really widely used in the Linux world, and it's become an essential tool for many developers. Git is also really well integrated with other Linux tools and technologies, such as GitHub and GitLab. These tools provide a really easy way to host and manage Git repositories, and they're widely used in the Linux community. Git is also really flexible, which makes it easy to use in all sorts of Linux projects. It's used by developers who are working on small scripts, and it's also used by developers who are working on huge operating

systems. Git is a really useful tool that's widely used in the Linux world, and its flexibility and ease of use make it a great choice for many developers. Git is also really well supported in the Linux community, with many online resources and tutorials available to help developers get started. This makes it really easy for new developers to learn Git and start using it in their own projects. Git is a great tool for Linux developers, and its wide use and flexibility make it a really valuable resource for the Linux community.

Part C — FOSS Ecosystem

The FOSS ecosystem around Git is really big and active. There are many other open source tools and technologies that are designed to work with Git, such as GitHub and GitLab. These tools provide a really easy way to host and manage Git repositories, and they're widely used in the open source community. There are also many other open source version control systems, such as Mercurial and Subversion. These systems are also widely used, and they provide a really alternative to Git for developers who need it. The FOSS ecosystem around Git is also really collaborative, with many developers working together to create new tools and technologies. This creates a really vibrant and dynamic community, where new ideas and innovations are always emerging. The FOSS ecosystem around Git is also really supportive, with many online resources and tutorials available to help developers get started. This makes it really easy for new developers to learn Git and start using it in their own projects. Git is a really useful tool that's widely used in the FOSS ecosystem, and its flexibility and ease of use make it a great choice for many developers. The FOSS ecosystem around Git is really big and active, and it's a great resource for developers who want to learn more about version control and open source software. Git is a great tool for FOSS developers, and its wide use and flexibility make it a really valuable resource for the FOSS community.

Part D — Open Source vs Proprietary

There are many proprietary alternatives to Git, such as Perforce and ClearCase. These systems are also widely used, and they provide a really alternative to Git for developers who need it. However, they're not as flexible or easy to use as Git, and they're often really expensive. This makes them less appealing to many developers, who prefer the freedom and flexibility of open source software. Git is also really well supported, with many online resources and tutorials available to help developers get started. This makes it really easy for new developers to learn Git and start using it in their own projects. The proprietary alternatives to Git are often less well supported, which can make them harder to use and less appealing to developers. Git is a really useful tool that's widely used by developers, and its flexibility and ease of use make it a great choice for many projects. The proprietary alternatives to Git are often less flexible and less easy to use, which makes them less appealing to many developers. Git is a great tool for developers who want a free and open source version control system, and its wide use and flexibility make it a really valuable resource for the development community. Git is also really well integrated with other open source tools and technologies, which makes it a great choice for many developers. The proprietary alternatives to Git are often less well integrated, which can make them less appealing to developers who want a seamless and integrated development experience.

Shell Scripts and Outputs

Script 1: 01-identify.sh

```
#!/bin/bash
#          Author:          Arjun          Tiwari                    (24BAI10141)

echo
"==============================================================================
=="
echo "                       Git AUDIT - SYSTEM IDENTITY                       "
echo
"==============================================================================
=="
echo          "Linux          Distribution:          $(lsb_release          -ds)"
echo      "Kernel      Version:                                $(uname      -r)"
echo      "Current      User:                                        $(whoami)"
echo        "Home        Directory:                                     $(pwd)"
echo      "System      Uptime:                                        $(uptime)"
echo             "Current            Date/Time:                         $(date)"
echo  "-----------------------------------------------------------------------
-------"
echo "Message: This system runs on Open Source software, providing freedom to
study,                 change,                 and                 distribute."
echo
"==============================================================================
=="
```

This script is pretty straightforward - it's designed to display some basic info about the system it's running on. When you run it, you'll see a bunch of details printed out in the terminal, including the Linux distribution, kernel version, current user, home directory, system uptime, and the current date and time. It's basically just a quick way to get a snapshot of what's going on with your system, and it also includes a message at the end about the system running on open source software. So, if you run this script, you can expect to see all this info displayed in a neat format, with some lines of text and separators to make it easy to read.

Script 2: 02-packages.sh

```
#!/bin/bash
#          Author:          Arjun          Tiwari                    (24BAI10141)

if          [          -f          "/etc/debian_version"          ];          then
  PACKAGE_MANAGER="apt"
  PACKAGE_NAME="git"
else
  PACKAGE_MANAGER="yum"
  PACKAGE_NAME="git"
fi

if   $PACKAGE_MANAGER   list   --installed   |   grep   -q   $PACKAGE_NAME;   then
  echo
"================================================================================
==="
  echo "                      Git AUDIT - PACKAGE INSPECTOR                   "
  echo
"================================================================================
==="
  echo "Status: $PACKAGE_NAME is INSTALLED on this $(lsb_release -ds) system."
  echo     "Version:     $(git     --version     |     awk     '{print     $3}')"
  echo  "-----------------------------------------------------------------------
----------"
  echo                  "FOSS                  Philosophy                  Notes:"
  echo  "  -  Git:  Git  is  a  free  and  open  source  distributed  version  control
system."
  echo  "  -  Linux:  Linux  is  a  free  and  open-source  operating  system."
  echo   "   -   Vim:   Vim   is   a   free   and   open-source   text   editor."
  echo   "   -   GCC:   GCC   is   a   free   and   open-source   compiler."
  echo
"================================================================================
==="
else
  echo     "$PACKAGE_NAME     is     NOT     installed     on     this     system."
fi
```

This script checks if the Git package is installed on your system, and if it is, it shows you the version and some info about free and open-source software. It first figures out what package manager your system is using, either apt or yum, and then checks if Git is installed. If Git is installed, it prints out a bunch of info, including the version of Git and some notes about open-source stuff like Linux and Vim. If Git isn't installed, it just says that Git isn't installed. So, if you run this script, you should see either a big block of text with Git version and other info, or a simple message saying Git isn't installed.

Script 3: 03-auditor.sh

```
#!/bin/bash
#          Author:          Arjun          Tiwari                    (24BAI10141)

dirs=(/etc         /var/log         /usr/bin         /usr/local/bin         /home)
dirs+=(/usr/share/git                                                 /var/lib/git)

echo
"================================================================================
=="
for               dir               in               "${dirs[@]}";               do
  if            [            -d            "$dir"            ];            then
    echo                                "$dir                                exists"
    echo     "Size:     $(du     -sh     $dir     |     awk     '{print     $1}')"
    echo         "Permissions:         $(stat         -c         %a         $dir)"
    echo          "Owner:          $(stat          -c          %U          $dir)"
    echo  "-----------------------------------------------------------------------
-----------"
  else
    echo              "$dir              does              not              exist"
  fi
done
echo
"================================================================================
=="
```

This script is pretty straightforward - it checks a list of directories on the system and reports back on whether they exist or not. If a directory does exist, it also shows the size of the directory, its permissions, and who owns it. The script will go through each directory in the list, which includes things like /etc, /var/log, and /home, and print out the results for each one. When you run it, you should see a bunch of output in the terminal that shows which directories are there, how big they are, what permissions they have, and who the owner is, with a separator line between each directory's info.

Script 4: 04-logs.sh

```
#!/bin/bash
#          Author:          Arjun          Tiwari                    (24BAI10141)

#    Suggest    a    realistic    log    path    for    Git:    /var/log/git.log
LOG_FILE=$1
KEYWORD=$2
echo
"================================================================================
==="
echo "                        Git AUDIT - LOG FILE ANALYZER                   "
echo
"================================================================================
==="
COUNT=0
while                read                -r                line;                do
  if        [[        $line        ==        *$KEYWORD*        ]];        then
    ((COUNT++))
  fi
done                                  <                                  $LOG_FILE
echo    "Keyword    '$KEYWORD'    found    $COUNT    times    in    $LOG_FILE"

tail        -n        5        $LOG_FILE        |        grep        $KEYWORD
echo
"================================================================================
==="
```

This script is pretty straightforward - it takes a log file and a keyword as input, then checks how many times that keyword shows up in the log file. It does this by reading the file line by line and incrementing a counter whenever it finds a match. After that, it prints out the total count of the keyword and the last 5 lines of the log file that contain the keyword. So, when you run the script, you should see a header with the script's name, the total count of the keyword, the last 5 relevant log lines, and a footer - that's the basic output you'd expect to see in the terminal.

Script 5: 05-manifesto.sh

```
#!/bin/bash
#          Author:          Arjun          Tiwari                    (24BAI10141)

echo
"================================================================================
=="
read        -p        "What        is        your        name?        "        NAME
read   -p   "What   is   your   favorite   open-source   project?   "   PROJECT
read  -p  "What  do  you  think  is  the  most  important  aspect  of  open-source
software?                                "                                ASPECT
echo
"================================================================================
=="

echo "My name is $NAME, and I believe in the power of open-source software. My
favorite project is $PROJECT, and I think the most important aspect of open-
source is $ASPECT. I will always support and contribute to the open-source
community."

echo
"================================================================================
=="

echo       "My       Open       Source       Manifesto"       >       $NAME.txt
echo "My name is $NAME, and I believe in the power of open-source software. My
favorite project is $PROJECT, and I think the most important aspect of open-
source is $ASPECT. I will always support and contribute to the open-source
community." >> $NAME.txt
```

This script is pretty straightforward - it asks the user for their name, favorite open-source project, and what they think is the most important aspect of open-source software. It then prints out a statement that includes the user's input, kind of like a personal manifesto. The same statement also gets written to a text file named after the user. So, when you run the script, you'll see a few prompts to enter your info, then it'll print out your manifesto in the terminal, and you'll also get a new text file with the same info.

Conclusion

In conclusion, Git is a really useful tool that's widely used by developers. It's free, open source, and really flexible, which makes it a great choice for many projects. Git is also really well supported, with many online resources and tutorials available to help developers get started. The open source license for Git is really important, because it ensures that Git remains a community-driven project. The community values behind Git are all about collaboration and openness, and the ethics behind Git are all about making it accessible and inclusive to everyone. Git has a really big footprint in the Linux world, and it's used by many other open source projects. The FOSS ecosystem around Git is really big and active, with many other open source tools and technologies that are designed to work with Git. There are many proprietary alternatives to Git, but they're often less flexible and less easy to use. Git is a great tool for developers who want a free and open source version control system, and its wide use and flexibility make it a really valuable resource for the development community. Overall, Git is a really useful tool that's widely used by developers, and its flexibility and ease of use make it a great choice for many projects. It's free, open source, and really well supported, which makes it a great resource for developers who want to learn more about version control and open source software. Git is a great choice for developers who want a free and open source version control system, and its wide use and flexibility make it a really valuable resource for the development community.